# Lie's Theorem for Solvable Complex Lie Algebras

## An End-to-End Meta-Math Walkthrough

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Abstract

An ideal of codimension one provides a weight vector by induction. A trace calculation prevents the remaining direction from changing that weight, enabling one last eigenvector selection. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**RT-04 / Representation Theory and Lie Theory / Report v1**

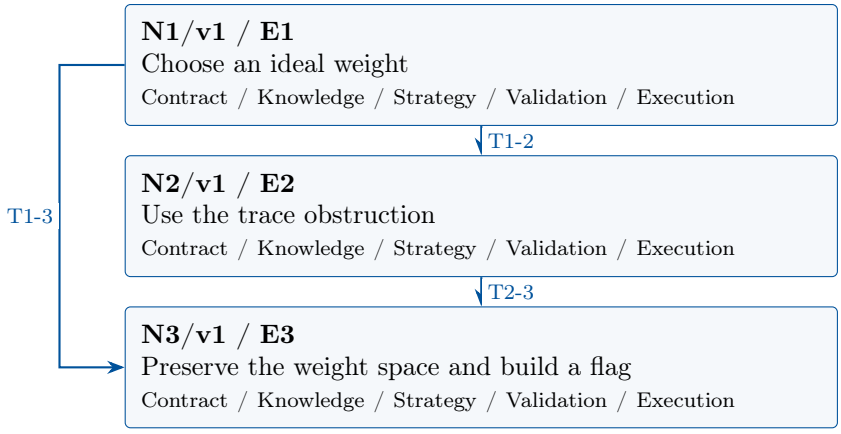
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *Let $\mathfrak{g}$ be a finite-dimensional solvable complex Lie algebra and $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ a representation on a nonzero finite-dimensional complex vector space. There is a common eigenvector: some $v \neq 0$ and linear $\lambda : \mathfrak{g} \to \mathbb{C}$ satisfy $\rho(y)v = \lambda(y)v$ for all $y$. Consequently there is a basis in which every $\rho(y)$ is upper triangular. Solvable means the iterated derived series eventually becomes zero.*

### A. Foundations and permitted imports

Use eigenvalue existence over $\mathbb{C}$, trace of a commutator equal to zero, induction on dimension, and the representation identity $\rho([x, y]) = [\rho(x), \rho(y)]$. Write $yv$ for $\rho(y)v$. A subalgebra of a solvable algebra is solvable. A nonzero solvable algebra has $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$, since equality would make its derived series constant and nonzero.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1-2: N1/E1/v1 to N2.** There exist $x$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero $v_0$, and a linear weight $\lambda$ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

**Receiving activity.** N2.1

**Compatibility.** E1/v1 and $W = \text{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Check: The finite span $W$ is invariant under both $x$ and $\mathfrak{h}$, so restricted commutator traces are legitimate.

**T1-3: N1/E1/v1 to N3.** There exist $x$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero $v_0$, and a linear weight $\lambda$ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

**Receiving activity.** N3.1

**Compatibility.** E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Check: $E \neq 0$ since it contains $v_0$; the final induction decreases representation dimension.

**T2-3: N2/E2/v1 to N3.** $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

**Receiving activity.** N3.1

**Compatibility.** E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Check: $E \neq 0$ since it contains $v_0$; the final induction decreases representation dimension.

## III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

### A. N1/v1: Choose an ideal weight

**Object and role.** Reduce the Lie algebra dimension, then record the orbit of a weight vector under the missing direction. **Local obligation:** There exist $x$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero $v_0$, and a linear weight $\lambda$ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

**Inputs.** Induction on $d = \dim \mathfrak{g}$ and a hyperplane $\mathfrak{h} \supset [\mathfrak{g}, \mathfrak{g}]$.

**C — Contract**

Publish E1/v1 only after establishing this obligation: There exist $x$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero $v_0$, and a linear weight $\lambda$ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

**K — Knowledge**

Induction on $d = \dim \mathfrak{g}$ and a hyperplane $\mathfrak{h} \supset [\mathfrak{g}, \mathfrak{g}]$. Retain the stated assumptions and import versions.

**S — Strategy**

Reduce the Lie algebra dimension, then record the orbit of a weight vector under the missing direction.

**V — Validation**

$\mathfrak{h}$ is an ideal because every bracket with it lies in $[\mathfrak{g}, \mathfrak{g}]$; induction decreases algebra dimension.

**E — Execution**

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N1.1 When $\mathfrak{g} = 0$, any nonzero vector works. For $d > 0$, solvability makes the derived algebra proper. Choose a hyperplane $\mathfrak{h}$ containing it and $x$ outside. The hyperplane is an ideal and solvable, of dimension $d - 1$.

N1.2 Apply the induction hypothesis to the restriction of the representation to $\mathfrak{h}$. It gives $v_0 \neq 0$ with $yv_0 = \lambda(y)v_0$. The scalar function $\lambda$ is linear because the representation is linear and $v_0$ is nonzero.

**Export E1/v1.** There exist $x$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero $v_0$, and a linear weight $\lambda$ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

**Acceptance gate.** $\mathfrak{h}$ is an ideal because every bracket with it lies in $[\mathfrak{g}, \mathfrak{g}]$; induction decreases algebra dimension.

**Targeted repair.** If an arbitrary hyperplane is assumed ideal, choose one containing the derived algebra before applying the commutator calculation. Withhold this export and recheck its descendants after a versioned correction.

### B. N2/v1: Use the trace obstruction

**Object and role.** Triangularize the ideal action on a finite cyclic span and take traces of commutators there. **Local obligation:** $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

**Inputs.** E1/v1 and $W = \operatorname{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Incoming proof exports: E1/v1 at N2.1.

**C — Contract**

Publish E2/v1 only after establishing this obligation: $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

**K — Knowledge**

E1/v1 and $W = \operatorname{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

**S — Strategy**

Triangularize the ideal action on a finite cyclic span and take traces of commutators there.

**V — Validation**

The finite span $W$ is invariant under both $x$ and $\mathfrak{h}$, so restricted commutator traces are legitimate.

**E — Execution**

Carry out activities N2.1–N2.3 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N2.1 Put $v_j = x^j v_0$. Induction on $j$, simultaneously for every $y \in \mathfrak{h}$, gives $yv_j - \lambda(y)v_j \in \operatorname{span}(v_0, \ldots, v_{j-1})$. Indeed $yxv_{j-1} = x(yv_{j-1}) + [y, x]v_{j-1}$; the first term has the required leading part and the second lies in the previous span because $[y, x] \in \mathfrak{h}$ and the induction hypothesis applies to it.

N2.2 Let $m \geq 1$ be the first index where $v_m$ depends on its predecessors. It exists by finite dimension. Then $v_0, \ldots, v_{m-1}$ form a basis of $W$, which is $x$-invariant and, by the preceding formula, $\mathfrak{h}$-invariant. For $y \in \mathfrak{h}$, its action on $W$ is triangular with all diagonal entries $\lambda(y)$.

N2.3 The action of $[x, y]$ on $W$ is the commutator of the two restricted operators, so its trace is zero. It also has trace $m\lambda([x, y])$, since $[x, y] \in \mathfrak{h}$. Characteristic zero gives $\lambda([x, y]) = 0$.

**Export E2/v1.** $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

**Acceptance gate.** The finite span $W$ is invariant under both $x$ and $\mathfrak{h}$, so restricted commutator traces are legitimate.

**Targeted repair.** If a trace is taken on a noninvariant partial span, first pass to the full finite cyclic space $W$ and establish both invariances. Withhold this export and recheck its descendants after a versioned correction.

### C. N3/v1: Preserve the weight space and build a flag

**Object and role.** Use the vanished commutator weights to show $xE \subset E$, then repeat on quotient spaces. **Local obligation:** A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

**Inputs.** E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

**C — Contract**

Publish E3/v1 only after establishing this obligation: A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

**K — Knowledge**

E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

**S — Strategy**

Use the vanished commutator weights to show $xE \subset E$, then repeat on quotient spaces.

**V — Validation**

$E \neq 0$ since it contains $v_0$; the final induction decreases representation dimension.

**E — Execution**

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N3.1 For $v \in E$ and $y \in \mathfrak{h}$, $y(xv) = x(yv) + [y, x]v = \lambda(y)xv + \lambda([y, x])v = \lambda(y)xv$. Thus $xE \subset E$. On this nonzero finite-dimensional complex space, $x$ has an eigenvector $v$. Since $\mathfrak{g} = \mathfrak{h} + \mathbb{C}x$, it is a common eigenvector for all of $\mathfrak{g}$, completing the induction on $d$.

N3.2 The line $\mathbb{C}v$ is invariant. To triangularize, induct separately on $\dim V$: the quotient $V/\mathbb{C}v$ has a representation of the same solvable algebra and a full invariant flag by induction. Lift its flag and prepend $\mathbb{C}v$. A basis adapted to this flag makes all representing matrices upper triangular.

**Export E3/v1.** A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

**Acceptance gate.** $E \neq 0$ since it contains $v_0$; the final induction decreases representation dimension.

**Targeted repair.** If $E$ is claimed invariant without E2, restore the commutator term and use its zero weight; otherwise the eigenvector argument is unsupported. Withhold this export and recheck its descendants after a versioned correction.

## IV. CONTINUOUS MATHEMATICAL PROOF

### A. N1 — Choose an ideal weight

1. **N1.1.** When $\mathfrak{g} = 0$, any nonzero vector works. For $d > 0$, solvability makes the derived algebra proper. Choose a hyperplane $\mathfrak{h}$ containing it and $x$ outside. The hyperplane is an ideal and solvable, of dimension $d - 1$.

2. **N1.2.** Apply the induction hypothesis to the restriction of the representation to $\mathfrak{h}$. It gives $v_0 \neq 0$ with $yv_0 = \lambda(y)v_0$. The scalar function $\lambda$ is linear because the representation is linear and $v_0$ is nonzero.

### B. N2 — Use the trace obstruction

1. **N2.1.** Put $v_j = x^j v_0$. Induction on $j$, simultaneously for every $y \in \mathfrak{h}$, gives $yv_j - \lambda(y)v_j \in \operatorname{span}(v_0, \ldots, v_{j-1})$. Indeed $yxv_{j-1} = x(yv_{j-1}) + [y, x]v_{j-1}$; the first term has the required leading part and the second lies in the previous span because $[y, x] \in \mathfrak{h}$ and the induction hypothesis applies to it.

2. **N2.2.** Let $m \geq 1$ be the first index where $v_m$ depends on its predecessors. It exists by finite dimension. Then $v_0, \ldots, v_{m-1}$ form a basis of $W$, which is $x$-invariant and, by the preceding formula, $\mathfrak{h}$-invariant. For $y \in \mathfrak{h}$, its action on $W$ is triangular with all diagonal entries $\lambda(y)$.

3. **N2.3.** The action of $[x, y]$ on $W$ is the commutator of the two restricted operators, so its trace is zero. It also has trace $m\lambda([x, y])$, since $[x, y] \in \mathfrak{h}$. Characteristic zero gives $\lambda([x, y]) = 0$.

### C. N3 — Preserve the weight space and build a flag

1. **N3.1.** For $v \in E$ and $y \in \mathfrak{h}$, $y(xv) = x(yv) + [y, x]v = \lambda(y)xv + \lambda([y, x])v = \lambda(y)xv$. Thus $xE \subset E$. On this nonzero finite-dimensional complex space, $x$ has an eigenvector $v$. Since $\mathfrak{g} = \mathfrak{h} + \mathbb{C}x$, it is a common eigenvector for all of $\mathfrak{g}$, completing the induction on $d$.

2. **N3.2.** The line $\mathbb{C}v$ is invariant. To triangularize, induct separately on $\dim V$: the quotient $V/\mathbb{C}v$ has a representation of the same solvable algebra and a full invariant flag by induction. Lift its flag and prepend $\mathbb{C}v$. A basis adapted to this flag makes all representing matrices upper triangular.

□

## V. WORKED ORCHESTRATION

Let $\mathfrak{g}$ be the upper-triangular $2 \times 2$ complex matrices acting on $\mathbb{C}^2$. Its derived algebra consists of strictly upper-triangular matrices, whose mutual brackets vanish, so it is solvable. The vector $e_1$ is a common eigenvector with weight $\lambda\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) = a$. The invariant flag is $0 \subset \mathbb{C}e_1 \subset \mathbb{C}^2$.

### A. Boundary and hypothesis audit

The standard representation of $\mathfrak{sl}_2(\mathbb{C})$ has no common eigenvector: the raising matrix kills only $\mathbb{C}e_1$, whereas the lowering matrix sends $e_1$ to $e_2$. This Lie algebra is not solvable. Over $\mathbb{R}$, even a one-dimensional abelian algebra can act by a rotation without a real eigenvector.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Rejected argument, diagnosis, and correction

**Rejected N3 step:** the $\mathfrak{h}$-weight space is automatically invariant under $x$. The identity $y(xv) = \lambda(y)xv + \lambda([y,x])v$ exposes the missing term. **Diagnosis:** Strategy omitted the trace obstruction. **Repair:** establish N2 on the invariant cyclic span, then substitute $\lambda([y,x]) = 0$. This fills a genuine logical gap under the original hypotheses; recheck the common eigenvector and flag exports.

### B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Compare this invariant-flag theorem with the highest-weight structure of nonsolvable $\mathfrak{sl}_2$ in RT-05.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.** Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.